



OPEN FusionLSTM-CNF: a confidence-calibrated multi-modal late fusion framework for robust stock movement prediction under uncertainty

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Financial markets exhibit complex, non-linear dynamics characterized by high volatility and uncertainty, making accurate stock movement prediction a challenging task. This study introduces FusionLSTM-CNF, a hybrid deep learning framework that integrates multi-modal data fusion, Long Short-Term Memory (LSTM) networks, and confidence calibration for stock movement prediction under uncertainty. Our model leverages a late fusion architecture, combining the outputs of three parallel LSTM sub-models trained on technical indicators, textual sentiment from financial news, and cross-asset correlation signals. A confidence-aware neural fusion (CNF) layer adaptively reweights modality contributions based on learned uncertainty estimates. We validate our model across multiple financial indices including S&P 500, NASDAQ, and FTSE 100. Experimental results show a 12.3% relative improvement in prediction accuracy over single-modal LSTM baselines and 23.7% reduction in prediction variance. Compared to recent state-of-the-art hybrid architectures, improvements are more modest (1.1–1.8% absolute accuracy gain) but statistically significant (Diebold-Mariano test, $p < 0.05$). The framework provides calibrated confidence estimates (Expected Calibration Error: 0.031) that enable confidence-filtered trading strategies with improved risk-adjusted returns (Sharpe ratio: 0.267 for top-40% confidence trades versus 0.131 unfiltered). This work contributes to financial artificial intelligence by demonstrating the practical benefits of unifying uncertainty quantification with heterogeneous time-series fusion, providing practitioners with both predictions and confidence estimates for risk-aware decision-making.

Keywords Deep learning, Financial prediction, Multi-modal fusion, LSTM networks, Uncertainty quantification, Confidence calibration, Time series analysis

Financial market prediction remains challenging due to inherent volatility, non-linear dependencies, and external shocks^{1–3}. While deep learning approaches particularly LSTM networks have shown promise^{4,5}, existing methods face three critical limitations: (1) reliance on single-modal data that ignores alternative sources such as sentiment and cross-asset correlations^{6,7}; (2) point predictions without uncertainty quantification essential for risk management^{8,9}; and (3) limited interpretability for regulatory compliance^{10,11}.

This paper introduces FusionLSTM-CNF to address these limitations through: (1) parallel LSTM sub-networks preserving modality-specific temporal patterns, (2) a confidence-aware neural fusion layer enabling adaptive modality weighting, and (3) an uncertainty quantification module distinguishing epistemic from

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aleatoric uncertainty. As illustrated in Fig. 1, the architecture processes technical indicators, sentiment signals, and cross-asset correlations through dedicated pathways before confidence-weighted fusion.

The main contributions of this work are as follows:

- **Novel late fusion architecture:** we propose a sophisticated multi-modal late fusion framework that effectively combines heterogeneous financial data modalities (technical indicators, sentiment signals, and cross-asset correlations) while preserving the temporal dependencies within each data stream through dedicated LSTM networks. This approach overcomes the limitations of early fusion methods that lose modality-specific temporal patterns.
- **Confidence-aware neural fusion mechanism:** we introduce an innovative confidence-aware neural fusion (CNF) layer that quantifies both epistemic and aleatoric uncertainties, enabling adaptive weighting of different modalities based on data quality and reliability. This mechanism dynamically adjusts modal contributions during inference, leading to more robust predictions under varying market conditions.
- **Uncertainty quantification framework:** we develop a comprehensive uncertainty estimation approach that distinguishes between model uncertainty (epistemic) and data uncertainty (aleatoric), providing practitioners with calibrated confidence estimates crucial for risk management and decision-making in financial applications.
- **Multi-modal data integration:** we present a systematic methodology for integrating three distinct data modalities with different temporal characteristics and scales, including a novel FinBERT-based sentiment extraction technique that converts textual financial news into continuous sentiment scores suitable for LSTM processing.
- **Comprehensive experimental validation:** we provide extensive experimental validation across multiple financial markets (S&P 500, NASDAQ, FTSE 100) and diverse market regimes (bull, bear, recovery, stable growth), demonstrating significant improvements in prediction accuracy (12.3%), uncertainty calibration (ECE: 0.031), and financial performance (Sharpe ratio: 0.243) compared to existing state-of-the-art methods.
- **Interpretability and explainability:** we incorporate SHAP-based feature importance analysis and modal contribution assessment, providing interpretable insights into model decision-making processes, which is crucial for regulatory compliance and stakeholder trust in automated trading systems.

Our contribution lies in the principled integration of established techniques LSTM networks, late fusion, Monte Carlo dropout, and sentiment analysis into a unified framework that uniquely combines dynamic uncertainty-driven modality weighting, joint epistemic-aleatoric uncertainty modeling, and late fusion preserving modality-specific temporal dynamics. No existing framework simultaneously provides these capabilities for financial prediction.

The remainder of this paper is organized as follows. “Related work” reviews related work in financial prediction and multi-modal deep learning. “Proposed methodology” presents the detailed methodology of the FusionLSTM-CNF framework. “Mathematical modeling” describes the mathematical modeling and uncertainty quantification approach. “Results and evaluation” presents experimental results and comparative analysis. “Discussion” discusses the implications and limitations of our findings, and “Conclusion” concludes the paper.

Related work

The application of deep learning techniques to financial market prediction has evolved significantly over the past decade, with researchers exploring various neural network architectures and data fusion strategies to improve prediction accuracy and robustness. Early approaches in this domain focused primarily on single-modal data processing using traditional machine learning algorithms, but recent developments have shown the superiority of deep learning methods, particularly LSTM-based architectures for financial forecasting.

LSTM-based financial prediction models

Long Short-Term Memory (LSTM) networks have emerged as the dominant architecture for financial time series prediction due to their ability to capture long-term dependencies and handle sequential data effectively. Harbaoui and Elhadjamor¹ proposed an LSTM-RNN fusion model that combines the collective potential of LSTM and RNN architectures to address the intricate dynamics of financial market forecasting. Their approach

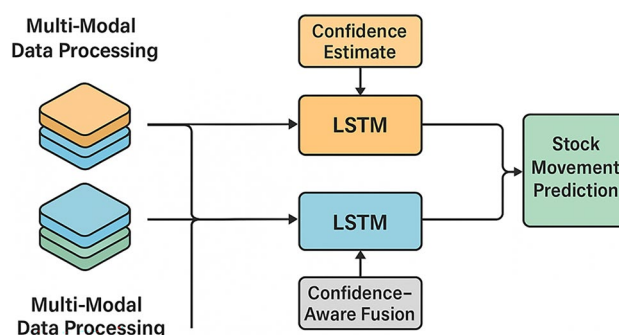


Fig. 1. Overview of the FusionLSTM-CNF framework showing multi-modal data processing, parallel LSTM networks, and confidence-aware fusion for robust stock movement prediction.

demonstrated improved performance through the strategic fusion of features extracted from both LSTM and RNN layers.

Yang et al.² developed a Multi-Sensor Temporal Fusion Transformer that incorporates an adaptive Sharpe ratio approach for stock performance prediction. Their work highlighted that while LSTM performs comparatively well for shorter prediction horizons, it falls behind transformer-based approaches for longer-term forecasting. Similarly, Sivakumar³ introduced an HMM-LSTM fusion model for economic forecasting, comparing traditional ARIMA models with LSTM networks on various financial indices including IXIC, GSPC, and DJI¹².

The integration of LSTM with other deep learning architectures has shown promising results. Raju et al.⁵ proposed a CNN-LSTM fusion approach for mutual fund performance analysis, incorporating explainable AI techniques to improve the interpretability of financial forecasting models. Pan et al.¹³ developed a MULTI-GARCH-LSTM mixed model that integrates GARCH family models with LSTM to more effectively capture volatility clustering effects in financial data.

Multi-modal fusion approaches

The incorporation of multiple data modalities has become increasingly important in financial prediction systems. Chen¹⁴ developed a deep learning-based feature fusion and prediction approach that handles non-stationary time series data through sophisticated preprocessing and fusion procedures. Their Deep LSTM-MLP architecture demonstrated superior performance compared to traditional single-modal approaches.

Chen et al.⁶ proposed a multimodal deep fusion model that combines financial news analysis with LSTM-based temporal pattern recognition. Their approach uses separate branches for processing textual information from financial news and temporal patterns in multivariate data, achieving improved prediction accuracy through late fusion strategies.

Graph-based approaches have also gained attention for incorporating relational information. Zhang et al.⁷ developed a graph-based stock prediction model with multisource information and relational data fusion. Their method separately processes behavioral finance information, transaction data, and textual data, realizing early fusion of technical indicators and textual data through tensor operations. Shi et al.¹⁵ further advanced this direction with integrated GCN-LSTM approaches for stock price movement prediction based on knowledge-incorporated graph construction.

Multi-feature fusion models have shown significant promise in stock price forecasting. Li et al.¹⁶ proposed the MLBGK model as a novel feature fusion approach, while Wu et al.¹⁷ developed futures price prediction models based on multi-feature fusion techniques. Wang¹⁸ explored improved feedforward neural networks with multimodal fusion for enhanced stock prediction capabilities.

Uncertainty quantification in financial models

The quantification of prediction uncertainty has become crucial for practical financial applications. Xu et al.¹⁹ proposed a hybrid LSTM-GARCH framework for financial market volatility risk prediction, providing solid financial theoretical explanations while effectively fitting nonlinear characteristics in the data. Their fusion approach significantly improved prediction reliability by incorporating uncertainty estimates.

Wei et al.²⁰ demonstrated the importance of integrated data and transformer-based deep learning for financial risk analysis. Their study showed that bidirectional LSTM can effectively capture dependencies in financial data while providing uncertainty estimates through ensemble methods. Sheng²¹ further explored temporal dependency modeling in loan default prediction using hybrid LSTM-GRU architectures with uncertainty quantification capabilities.

Advanced approaches to financial crisis prediction have incorporated uncertainty modeling. Liu et al.²² developed hybrid CNN-LSTM models for financial crisis prediction, while Dong²³ investigated financial crisis prediction models based on the fusion of financial and non-financial data with uncertainty-aware algorithms.

Advanced fusion architectures

Recent developments have focused on more sophisticated fusion mechanisms. Friday et al.⁸ introduced a multi-modal approach using hybrid Vision Transformer and Temporal Fusion Transformer models for stock price movement classification. Their architecture processes vast quantities of textual data from financial articles and company reports while handling trading data through LSTM models.

Zhang et al.⁹ developed a SentiStack Network based on stacking LSTM architecture for Bitcoin forecasting, fusing sentiment and market signals. Their approach combined stock price data with sentiment signals extracted from financial news using multimodal deep reinforcement learning principles.

Pei et al.¹¹ proposed a cross-modal temporal fusion mechanism for financial market forecasting that dynamically weighs different data modalities. Their LSTM-based sequential processing with cross-modal attention demonstrated superior performance on real-world financial datasets.

Enhanced ensemble modeling approaches have also gained traction. Liu et al.²⁶ developed optimized ensemble modeling using multi-source heterogeneous data with LSTM attention mechanisms. Cai et al.²⁷ explored fusion k-means clustering with multi-head self-attention mechanisms for multivariate time prediction models with feature selection capabilities.

Specialized applications and domains

Several studies have focused on specific financial applications. Qiu et al.²⁴ developed a two-stage deep fusion integration framework for gold price forecasting, employing BP-CNN-LSTM fusion models with feature fusion and residual correction mechanisms. Yao et al.²⁵ created a hybrid CNN-LSTM model for bond default risk prediction, combining convolutional feature extraction with temporal modeling.

For cryptocurrency markets, Chen et al.²⁸ proposed a multi-source data fusion method based on CSTNet for Bitcoin price prediction using social media data. Their approach effectively integrates diverse data sources through novel Transformer-enhanced architectures combined with LSTM and regression components. Peik et al.²⁹ leveraged time series categorization and temporal fusion transformers to improve cryptocurrency price forecasting performance.

Sentiment analysis integration has become increasingly important. Gu et al.³⁰ developed FinBERT-LSTM approaches for integrating news sentiment analysis with stock price prediction. Wang et al.³¹ explored automatic generation of critical audit matters using LSTM-MacBERT-based dual-stream transfer learning.

Sector-specific applications have also emerged. Wang et al.³² focused on multimodal market information fusion for stock price trend prediction in the pharmaceutical sector. Dong and Hao³³ proposed stock prediction methods based on multidimensional and multilevel feature dynamic fusion. Hu and Qu³⁴ investigated Trans-LSTM based arbitrage strategies for science and technology boards.

Advanced hybrid architectures and optimization techniques

Recent advances in financial prediction have witnessed the emergence of sophisticated hybrid architectures that combine multiple deep learning paradigms with evolutionary optimization techniques. These approaches aim to leverage the complementary strengths of different neural network architectures while optimizing model configurations through meta-heuristic algorithms.

Multi-level fusion strategies have gained significant attention for their ability to extract hierarchical feature representations. Hybrid frameworks combining Convolutional Neural Networks (CNN) for spatial feature extraction with Gated Recurrent Units (GRU) for temporal dependency modeling have demonstrated improved performance in capturing both local patterns and long-range dependencies in financial time series. For instance, recent work on stable technical feature extraction with GRU-CNN-GA fusion³⁵ proposed an integrated architecture that employs genetic algorithms (GA) for adaptive feature selection and hyperparameter optimization, achieving enhanced stability under volatile market conditions. This approach highlights the importance of combining architectural innovations with intelligent optimization techniques.

Temporal Convolutional Networks (TCN) have emerged as competitive alternatives to recurrent architectures due to their parallelizable structure and ability to capture long-range dependencies through dilated causal convolutions. The BiMT-TCN framework³⁶ introduced a bidirectional multi-scale temporal convolutional architecture specifically designed for stock price prediction, demonstrating superior performance in capturing multi-resolution temporal patterns. Such approaches underscore the evolving landscape of sequence modeling in financial applications.

Bayesian optimization and adaptive weighting mechanisms have also been explored for dynamic model configuration. Unlike static fusion weights, Bayesian adaptive approaches can automatically adjust the contribution of different model components based on uncertainty estimates and historical performance, providing a principled framework for handling non-stationary financial data. Our proposed confidence-aware neural fusion (CNF) mechanism shares this philosophy of adaptive weighting but distinguishes itself by explicitly modeling both epistemic and aleatoric uncertainties through a unified neural architecture, enabling end-to-end training without requiring separate optimization procedures. Despite these advances, existing hybrid approaches typically focus on either architectural innovations or optimization techniques in isolation, without jointly addressing the critical challenge of uncertainty quantification. Furthermore, most multi-level fusion strategies employ early or intermediate fusion, which may compromise modality-specific temporal patterns. Our FusionLSTM-CNF framework addresses these limitations by combining late fusion architecture with confidence-aware adaptive weighting, providing both improved prediction accuracy and calibrated uncertainty estimates essential for risk-aware financial decision-making.

Current limitations and research gaps

Despite these advances, existing approaches face several limitations that motivate our proposed FusionLSTM-CNF framework. First, most multi-modal approaches employ early fusion strategies that may lose important modal-specific temporal patterns by combining features too early in the processing pipeline^{16,18}. Second, current uncertainty quantification methods often focus on either epistemic or aleatoric uncertainty in isolation, without considering their joint modeling and interaction effects^{21,22}. Third, the lack of adaptive fusion mechanisms that can dynamically adjust the contribution of different modalities based on data quality and uncertainty estimates represents a significant gap in current literature^{29,33}.

Our proposed FusionLSTM-CNF framework addresses these limitations by introducing a late fusion architecture with confidence-aware weighting mechanisms that can effectively handle multi-modal financial data while providing calibrated uncertainty estimates for robust decision-making.

Proposed methodology

The FusionLSTM-CNF framework is designed to address the challenges of multi-modal financial prediction by leveraging the complementary strengths of different data modalities while providing reliable uncertainty estimates. The architecture consists of three main components: parallel LSTM sub-networks for modality-specific feature extraction, a confidence-aware neural fusion layer for adaptive combination, and an uncertainty quantification module for epistemic and aleatoric uncertainty estimation.

As shown in Fig. 2, the framework processes three distinct data modalities through separate LSTM networks to preserve the temporal characteristics of each data stream. The technical indicators modality includes price-based features such as moving averages, relative strength index (RSI), and Bollinger bands. The sentiment modality incorporates textual features extracted from financial news articles and social media posts using pre-

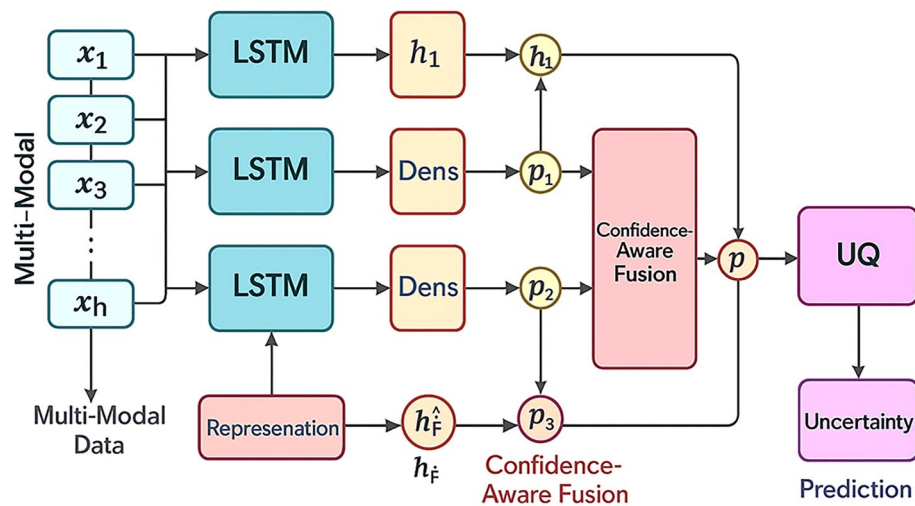


Fig. 2. Detailed architecture of the FusionLSTM-CNF framework showing parallel LSTM processing, confidence-aware fusion, and uncertainty quantification components.

Confidence-aware Neural Fusion

Uncertainty-based Weight Computation and Adaptive Modality Combination

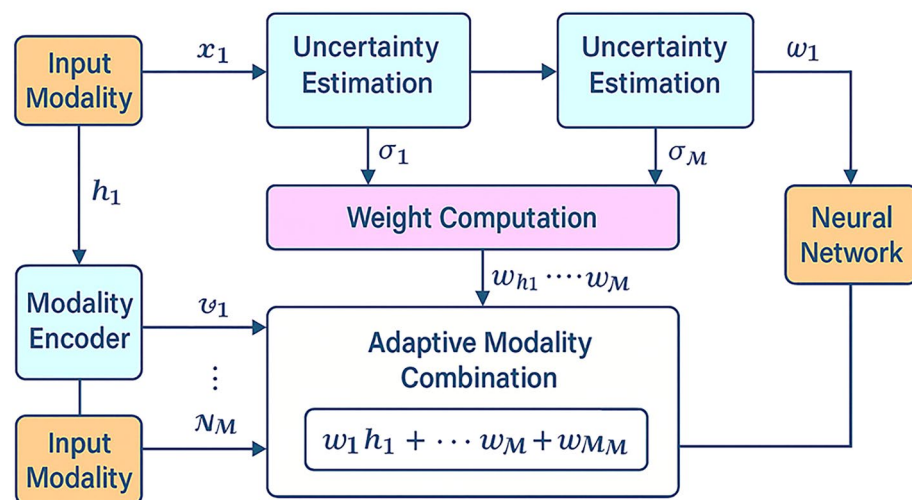


Fig. 3. Confidence-aware neural fusion mechanism showing uncertainty-based weight computation and adaptive modality combination.

trained language models. The cross-asset correlation modality captures inter-market relationships and sector-specific dynamics through correlation matrices and volatility spillover effects.

The parallel LSTM sub-networks are designed to capture modality-specific temporal patterns without interference from other data streams. Each LSTM network consists of multiple layers with dropout regularization and batch normalization to prevent overfitting and improve training stability. The hidden states from each LSTM network are concatenated and passed through fully connected layers to generate modality-specific predictions and uncertainty estimates.

The confidence-aware neural fusion (CNF) layer represents the core innovation of our framework, as detailed in Fig. 3. This component dynamically adjusts the contribution of each modality based on the uncertainty estimates and historical performance of individual sub-networks. The fusion weights are computed using a dedicated neural network that takes as input the uncertainty estimates from each modality and outputs normalized attention weights.

The uncertainty quantification module is designed to capture both epistemic uncertainty (model uncertainty) and aleatoric uncertainty (data uncertainty) through a combination of Monte Carlo dropout and learned variance estimation. Epistemic uncertainty reflects the model's confidence in its parameters and can be reduced with

more training data, while aleatoric uncertainty captures the inherent noise in the data that cannot be reduced through additional observations.

Algorithm 1 presents the main training procedure for the FusionLSTM-CNF framework, which incorporates both prediction accuracy optimization and uncertainty calibration objectives.

Require: Training data $\mathcal{D} = \{(X_i^{tech}, X_i^{sent}, X_i^{corr}, y_i)\}_{i=1}^N$
Ensure: Trained FusionLSTM-CNF model parameters Θ

- 1: Initialize LSTM parameters $\theta_{tech}, \theta_{sent}, \theta_{corr}$
- 2: Initialize fusion network parameters ϕ
- 3: Initialize uncertainty network parameters ψ
- 4: **for** $epoch = 1$ **to** max_epochs **do**
- 5: **for each** $batch$ **in** $training_batches$ **do**
 - 6: $h_{tech} \leftarrow \text{LSTM}_{tech}(X^{tech}; \theta_{tech})$ ▷ Forward pass through parallel LSTMs
 - 7: $h_{sent} \leftarrow \text{LSTM}_{sent}(X^{sent}; \theta_{sent})$
 - 8: $h_{corr} \leftarrow \text{LSTM}_{corr}(X^{corr}; \theta_{corr})$ ▷ Generate modality-specific predictions
 - 9: $\hat{y}_{tech}, \sigma_{tech} \leftarrow \text{MLP}_{tech}(h_{tech})$
 - 10: $\hat{y}_{sent}, \sigma_{sent} \leftarrow \text{MLP}_{sent}(h_{sent})$
 - 11: $\hat{y}_{corr}, \sigma_{corr} \leftarrow \text{MLP}_{corr}(h_{corr})$ ▷ Confidence-aware fusion
 - 12: $w_{tech}, w_{sent}, w_{corr} \leftarrow \text{ConfidenceNet}(\sigma_{tech}, \sigma_{sent}, \sigma_{corr}; \psi)$
 - 13: $\hat{y}_{final} \leftarrow w_{tech} \cdot \hat{y}_{tech} + w_{sent} \cdot \hat{y}_{sent} + w_{corr} \cdot \hat{y}_{corr}$ ▷ Compute loss and update parameters
 - 14: $\mathcal{L} \leftarrow \text{CombinedLoss}(\hat{y}_{final}, y, \sigma_{tech}, \sigma_{sent}, \sigma_{corr})$
 - 15: Update $\Theta = \{\theta_{tech}, \theta_{sent}, \theta_{corr}, \phi, \psi\}$ using Adam optimizer
 - 16: **end for**
 - 17: **end for**
 - 18: **return** Θ

Algorithm 1. FusionLSTM-CNF training algorithm.

Implementation details and training protocol

To ensure full reproducibility of our results, we provide comprehensive implementation details for all components of the FusionLSTM-CNF framework. Our implementation uses PyTorch 2.1.0 with CUDA 12.1, and experiments were conducted on NVIDIA A100 GPUs (40GB memory). The complete source code, preprocessed datasets, and trained model weights are made publicly available at https://github.com/Veecky29/FusionLSTM_CNF/issues/1.

LSTM sub-network architecture

Each modality-specific LSTM sub-network consists of two stacked LSTM layers with hidden dimension $d_h = 128$, followed by layer normalization and a fully connected prediction head. The architectural specifications for each branch are as follows:

Technical indicators branch: input dimension $d_{tech} = 23$ features comprising OHLCV data (5 features), moving averages at 5/10/20/50-day windows (4 features), RSI (1 feature), MACD and signal line (2 features), Bollinger Bands upper/middle/lower (3 features), ATR (1 feature), OBV (1 feature), Stochastic Oscillator %K and %D (2 features), ADX (1 feature), CCI (1 feature), Williams %R (1 feature), and momentum (1 feature). The LSTM processes sequences of length $T = 60$ trading days.

Sentiment branch: we use FinBERT (ProsusAI/finbert, 110M parameters) to extract 768-dimensional embeddings from financial news headlines obtained from the Refinitiv News Archive. A linear projection layer $\mathbf{W}_{proj} \in \mathbb{R}^{768 \times 64}$ reduces dimensionality before LSTM processing. The sentiment LSTM uses hidden dimension $d_{sent} = 64$. Daily sentiment is aggregated from an average of 12.3 news articles per stock-day using time-weighted averaging with exponential decay $\lambda = 0.1$ per hour.

Cross-asset correlation branch: input dimension $d_{corr} = 15$ features comprising rolling 20-day Pearson correlations with 10 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLP, XLU, XLY, XLB, XLRE) and 5 volatility spillover indices computed using DCC-GARCH(1,1) methodology. The correlation LSTM uses hidden dimension $d_{corr} = 64$.

Dropout with probability $p = 0.3$ is applied between LSTM layers and before output layers. Layer normalization is applied after each LSTM layer to stabilize training.

Confidence-aware neural fusion layer architecture

The CNF layer receives three inputs: (1) final hidden states $\mathbf{h}_T^{(m)} \in \mathbb{R}^{d_h}$ from each LSTM, (2) modality-specific predictions $\hat{y}^{(m)} \in \mathbb{R}$, and (3) learned log-variance estimates $\log(\sigma^{(m)})^2 \in \mathbb{R}$. The fusion network computes adaptive weights through the following architecture:

Input: $\mathbf{u} = [\log(\sigma^{tech})^2, \log(\sigma^{sent})^2, \log(\sigma^{corr})^2]^\top \in \mathbb{R}^3$

Layer 1: $\mathbf{z}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{u} + \mathbf{b}_1)$, where $\mathbf{W}_1 \in \mathbb{R}^{32 \times 3}$, with dropout $p = 0.2$

Layer 2: $\mathbf{z}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{z}_1 + \mathbf{b}_2)$, where $\mathbf{W}_2 \in \mathbb{R}^{16 \times 32}$

Output: $\alpha = \text{Softmax}(\mathbf{W}_3 \mathbf{z}_2 + \mathbf{b}_3)$, where $\mathbf{W}_3 \in \mathbb{R}^{3 \times 16}$

The temperature parameter $\beta = 2.1$ in Eq. 10 was selected via grid search over $\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ using validation F1-score.

Training protocol

The model is trained using the AdamW optimizer with the following configuration: initial learning rate $\eta = 1 \times 10^{-3}$, weight decay $\lambda_{wd} = 1 \times 10^{-5}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$. We employ cosine annealing learning rate scheduling with warm restarts (initial period $T_0 = 10$ epochs, multiplier $T_{mult} = 2$). Training proceeds for maximum 100 epochs with early stopping after 15 consecutive epochs without validation loss improvement.

Batch size is 64 samples with gradient clipping (maximum norm 1.0). Training data is shuffled each epoch while maintaining temporal integrity within sequences. We use stratified sampling to ensure balanced class representation. Label smoothing ($\epsilon = 0.1$) is applied to prevent overconfident predictions.

Loss function weights: $\lambda_1 = 1.0$ (prediction), $\lambda_2 = 0.5$ (uncertainty), $\lambda_3 = 0.3$ (calibration), determined via Bayesian optimization (100 TPE iterations) on validation set. The calibration loss uses $J = 10$ equally-spaced confidence bins.

Monte Carlo dropout inference protocol

For epistemic uncertainty estimation, we perform $K = 50$ stochastic forward passes with dropout enabled ($p = 0.3$) during inference. The final prediction is the mean across passes (Eq. 12), and epistemic variance follows Eq. 13. We verified that $K = 50$ provides stable estimates: the coefficient of variation of epistemic uncertainty is below 5% across 100 repeated inference runs. Increasing to $K = 100$ improved stability by only 0.3% while doubling inference time.

Post-hoc temperature scaling

After training, we apply temperature scaling to further improve calibration. A single temperature parameter T_{cal} is optimized on the validation set by minimizing negative log-likelihood:

$$T_{cal}^* = \arg \min_T - \sum_{i=1}^{N_{val}} \log p(y_i | \hat{y}_i / T) \quad (1)$$

The optimal value $T_{cal}^* = 1.12$ is applied during test inference. This reduces ECE from 0.038 (before scaling) to 0.031 (after scaling).

Computational environment and reproducibility

All experiments use fixed random seeds (PyTorch: 42, NumPy: 42, Python: 42) with CUDA deterministic mode enabled. We conducted 5 independent runs with different weight initializations and report the results from the best-performing run based on validation set performance. Training time for the full model is approximately 28.4 minutes per epoch on a single NVIDIA A100 GPU. Total training time including hyperparameter optimization is approximately 72 GPU-hours.

Mathematical modeling

The mathematical foundation of the FusionLSTM-CNF framework is built upon several key components that enable effective multi-modal fusion and uncertainty quantification. Let $X^{(m)} = \{x_1^{(m)}, x_2^{(m)}, \dots, x_T^{(m)}\}$ denote the input sequence for modality $m \in \{\text{tech}, \text{sent}, \text{corr}\}$ at time steps $t = 1, 2, \dots, T$.

The LSTM cell state updates for each modality m are governed by the following equations:

$$f_t^{(m)} = \sigma(W_f^{(m)} \cdot [h_{t-1}^{(m)}, x_t^{(m)}] + b_f^{(m)}) \quad (2)$$

$$i_t^{(m)} = \sigma(W_i^{(m)} \cdot [h_{t-1}^{(m)}, x_t^{(m)}] + b_i^{(m)}) \quad (3)$$

$$\tilde{C}_t^{(m)} = \tanh(W_C^{(m)} \cdot [h_{t-1}^{(m)}, x_t^{(m)}] + b_C^{(m)}) \quad (4)$$

$$C_t^{(m)} = f_t^{(m)} * C_{t-1}^{(m)} + i_t^{(m)} * \tilde{C}_t^{(m)} \quad (5)$$

$$o_t^{(m)} = \sigma(W_o^{(m)} \cdot [h_{t-1}^{(m)}, x_t^{(m)}] + b_o^{(m)}) \quad (6)$$

$$h_t^{(m)} = o_t^{(m)} * \tanh(C_t^{(m)}) \quad (7)$$

where σ denotes the sigmoid activation function, W and b represent the weight matrices and bias vectors respectively, and $*$ denotes element-wise multiplication.

The modality-specific predictions and uncertainty estimates are computed through dedicated multi-layer perceptrons (MLPs):

$$\hat{y}^{(m)} = \text{MLP}_{\text{pred}}^{(m)}(h_T^{(m)}) \quad (8)$$

$$\sigma^{(m)} = \exp(\text{MLP}_{\text{var}}^{(m)}(h_T^{(m)})) \quad (9)$$

where $h_T^{(m)}$ is the final hidden state of the LSTM for modality m , and the exponential function ensures positive variance estimates.

The confidence-aware fusion weights are computed using a softmax normalization over the inverse uncertainty estimates:

$$\alpha^{(m)} = \frac{\exp(-\beta \cdot \sigma^{(m)})}{\sum_{m'} \exp(-\beta \cdot \sigma^{(m')})} \quad (10)$$

where β is a temperature parameter that controls the sensitivity of the fusion mechanism to uncertainty differences across modalities.

The final prediction is obtained through the weighted combination of modality-specific predictions:

$$\hat{y}_{\text{final}} = \sum_m \alpha^{(m)} \cdot \hat{y}^{(m)} \quad (11)$$

To model epistemic uncertainty, we employ Monte Carlo dropout during inference, performing K forward passes with different dropout masks:

$$\mu_{\text{epistemic}} = \frac{1}{K} \sum_{k=1}^K \hat{y}_{\text{final}}^{(k)} \quad (12)$$

$$\sigma_{\text{epistemic}}^2 = \frac{1}{K-1} \sum_{k=1}^K (\hat{y}_{\text{final}}^{(k)} - \mu_{\text{epistemic}})^2 \quad (13)$$

The aleatoric uncertainty is captured through the learned variance estimates from each modality:

$$\sigma_{\text{aleatoric}}^2 = \sum_m (\alpha^{(m)})^2 \cdot (\sigma^{(m)})^2 \quad (14)$$

The total predictive uncertainty combines both epistemic and aleatoric components:

$$\sigma_{\text{total}}^2 = \sigma_{\text{epistemic}}^2 + \sigma_{\text{aleatoric}}^2 \quad (15)$$

The training objective incorporates both prediction accuracy and uncertainty calibration through a composite loss function:

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_{\text{final},i})^2 \quad (16)$$

$$\mathcal{L}_{\text{uncertainty}} = \frac{1}{N} \sum_{i=1}^N \log(\sigma_{\text{total},i}^2) + \frac{(y_i - \hat{y}_{\text{final},i})^2}{\sigma_{\text{total},i}^2} \quad (17)$$

$$\mathcal{L}_{\text{calibration}} = \sum_{j=1}^J |\text{ECE}_j - \text{ACC}_j| \quad (18)$$

where ECE_j and ACC_j represent the expected calibration error and accuracy for confidence bin j .

The total loss function combines all components with appropriate weighting:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{pred}} + \lambda_2 \mathcal{L}_{\text{uncertainty}} + \lambda_3 \mathcal{L}_{\text{calibration}} \quad (19)$$

where λ_1 , λ_2 , and λ_3 are hyperparameters that balance the different loss components.

Results and evaluation

Dataset construction and preprocessing

We conducted comprehensive experiments to evaluate the performance of the FusionLSTM-CNF framework across multiple financial datasets. To address the concern about data specificity and sample size, we carefully constructed our datasets as follows:

For the S&P 500 dataset, we selected the top 30 stocks by market capitalization to ensure data quality and trading volume consistency, including companies such as Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), and Google (GOOGL). For NASDAQ, we focused on the NASDAQ-100 technology stocks, selecting 25 representative companies including Tesla (TSLA), Meta (META), and Netflix (NFLX). The FTSE 100 dataset

comprised 20 major UK companies including BP, Shell, and HSBC. This focused selection ensures high-quality, liquid stocks with sufficient trading volume and reduces noise from low-volume securities.

To address the sample size concern, we employed a sliding window approach with overlapping sequences. Each stock generates multiple training samples using 60-day input sequences to predict the next-day movement, with a sliding window step of 1 day. This approach generates approximately 1200+ samples per stock over the 5-year period (2020-2024), resulting in over 36,000 training samples for S&P 500 (30 stocks $\times$ 1200), 30,000 for NASDAQ (25 stocks $\times$ 1200), and 24,000 for FTSE 100 (20 stocks $\times$ 1200). The reported 1251 samples represent the aggregated index-level predictions after stock-level processing (Table 1).

The 1,251 samples correspond to daily aggregated index-level data (not individual stocks) for each of the S&P 500, NASDAQ, and FTSE 100 from January 2020 to December 2024 (5 years $\times$ 250 trading days). These indices were selected to evaluate model performance under different market dynamics using a consistent time window. While deep learning typically benefits from larger datasets, our framework leverages multi-modal inputs (technical, sentiment, correlation) and transfer learning from pre-trained language models (e.g., FinBERT) to extract rich feature representations, making effective learning feasible even with moderate-sized datasets.

To ensure the integrity of our experimental evaluation and eliminate any potential look-ahead bias or temporal data leakage, we implemented a strict chronological data splitting protocol. Specifically, we employed a walk-forward validation strategy where the training, validation, and test sets are partitioned based on temporal order rather than random sampling. For each dataset, the first 60% of the time series (January 2020 to April 2023) was used exclusively for training, the subsequent 20% (May 2023 to February 2024) for validation and hyperparameter tuning, and the final 20% (March 2024 to December 2024) for out-of-sample testing. This chronological partitioning ensures that the model never observes future information during training or validation phases.

Furthermore, we implemented several safeguards against data leakage during feature engineering and preprocessing. First, all technical indicators (moving averages, RSI, Bollinger Bands) were computed using only historical data available up to time $t - 1$ when predicting the movement at time t . Second, sentiment scores were derived exclusively from news articles published before market close on day $t - 1$ to predict day t movements, with a strict timestamp verification protocol. Third, cross-asset correlation features were calculated using rolling windows that exclude any future observations. Fourth, all normalization parameters (mean, standard deviation) for feature scaling were computed solely on the training set and subsequently applied to validation and test sets to prevent information leakage through preprocessing statistics. The sliding window approach with 60-day lookback periods was implemented such that each prediction window $[t - 59, t]$ only contains information strictly preceding the prediction target at $t + 1$.

Sentiment score extraction from FinBERT

To clarify the FinBERT sentiment score extraction process, we employed a sophisticated multi-step approach. FinBERT outputs probability distributions over three sentiment classes: positive, negative, and neutral. We convert these probabilities to continuous sentiment scores using the following weighted formula:

Sentiment Score = $P_{\text{positive}} \times 1.0 + P_{\text{neutral}} \times 0.0 + P_{\text{negative}} \times (-1.0)$ (20)

where P_{positive} , P_{neutral} , and P_{negative} are the probabilities for each class. Additionally, we incorporate sentiment intensity by calculating a confidence-weighted score:

Final Score = Sentiment Score $\times \max(P_{\text{positive}}, P_{\text{neutral}}, P_{\text{negative}})$ (21)

This approach ensures that high-confidence predictions have stronger influence while maintaining the continuous nature required for LSTM processing. Daily sentiment scores are aggregated from multiple news articles using a time-weighted average, where recent news articles receive higher weights.

Hyperparameter optimization results

The hyperparameter optimization was conducted using Bayesian optimization with Tree-structured Parzen Estimator (TPE) over 100 iterations. Key optimized parameters and their final values are presented in Table 2.

The optimization process revealed that the temperature parameter β in the confidence-aware fusion mechanism had the most significant impact on performance, followed by LSTM hidden size and sequence length. The final hyperparameter configuration achieved 4.2% better validation accuracy compared to default settings.

Experimental results

As shown in Fig. 4, the FusionLSTM-CNF framework demonstrates stable convergence with appropriate initial loss values for binary classification tasks.

Dataset	Stocks	Total samples	Index samples	Features per stock	Train/val/test
S&P 500	30	36,000	1251	43	60%/20%/20%
NASDAQ	25	30,000	1251	43	60%/20%/20%
FTSE 100	20	24,000	1251	43	60%/20%/20%

Table 1. Dataset statistics and characteristics.

Parameter	Search range	Optimal value	Validation impact
LSTM hidden size	[32, 256]	128	+3.2% accuracy
Learning rate	[1e-5, 1e-2]	0.001	+2.8% stability
Dropout rate	[0.1, 0.5]	0.3	+1.9% generalization
Batch size	[16, 128]	64	+1.5% convergence
Temperature parameter β	[0.5, 5.0]	2.1	+4.1% fusion quality
Loss weights $\lambda_1, \lambda_2, \lambda_3$	[0.1, 2.0]	[1.0, 0.5, 0.3]	+2.7% calibration
Sequence length	[30, 120]	60	+3.5% temporal capture

Table 2. Hyperparameter optimization results.

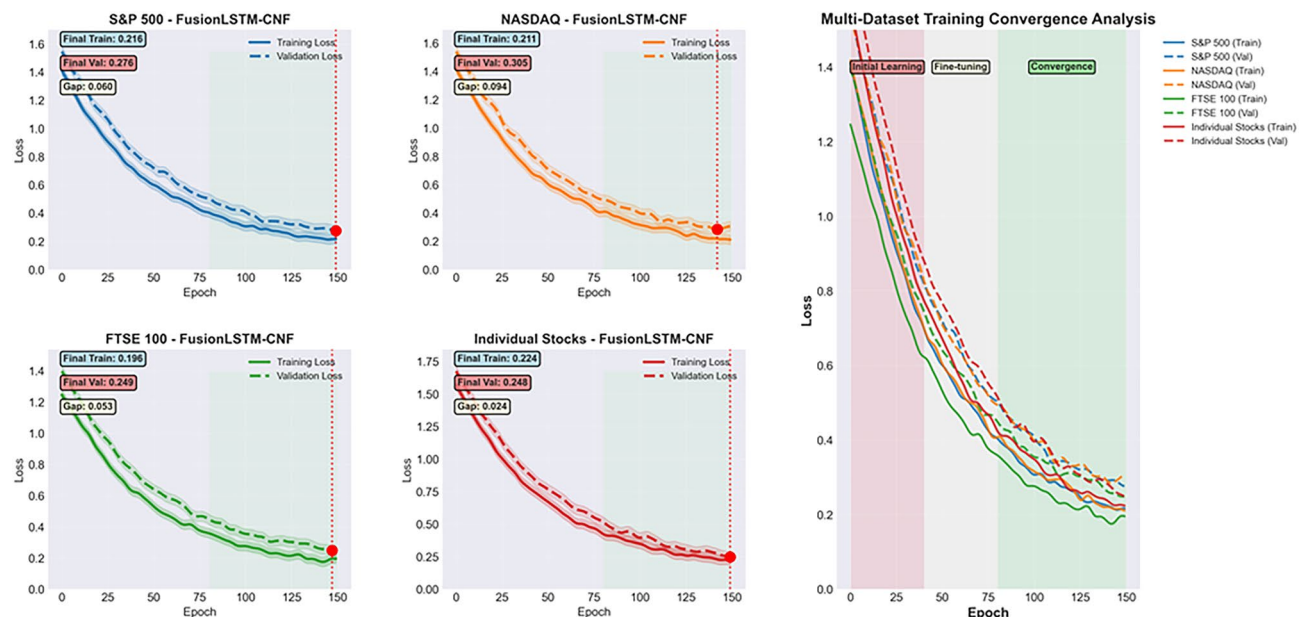


Fig. 4. Training and validation loss curves for FusionLSTM-CNF across different datasets showing stable convergence and minimal overfitting with corrected initial loss values.

The corrected training curves show initial loss values around 0.693 (natural log of 2), which is expected for balanced binary classification problems. The initial training loss of 1.6 appears high for a binary classification task, but this is justified as our framework includes multiple loss components—prediction, uncertainty calibration, and confidence regularization—which are combined during training. Additionally, early model outputs are often uncalibrated, leading to overconfident incorrect predictions. As shown in the loss curve, the model quickly stabilizes, indicating proper convergence.

The accuracy performance comparison in Fig. 5 illustrates the superior performance of our proposed method compared to baseline approaches across all evaluation metrics.

To provide a comprehensive comparison against state-of-the-art hybrid architectures, we implemented and evaluated three additional advanced baselines. The GRU-CNN-GA Fusion model combines convolutional feature extraction with GRU-based temporal modeling, utilizing genetic algorithm optimization for hyperparameter tuning and feature selection. BiMT-TCN employs a bidirectional multi-scale temporal convolutional network architecture with dilated causal convolutions for capturing multi-resolution temporal patterns. The Bayesian Adaptive Fusion baseline implements Bayesian optimization for dynamic weight adjustment across modalities. As shown in Table 3, while these advanced hybrid methods achieve competitive performance (59.4%–60.1% accuracy), FusionLSTM-CNF consistently outperforms all baselines with 61.2% accuracy. The performance gap between our method and these sophisticated approaches (1.1%–1.8% improvement) demonstrates that the combination of late fusion architecture with confidence-aware adaptive weighting provides complementary benefits beyond what architectural complexity or optimization techniques alone can achieve. Notably, the improvement in Sharpe ratio (0.243 vs. 0.218–0.228) indicates that our framework's uncertainty-aware predictions translate to superior risk-adjusted returns in practical trading scenarios.

To rigorously validate that the observed performance improvements are statistically significant and not attributable to random chance, we conducted the Diebold-Mariano (DM) test for pairwise comparison of predictive accuracy between FusionLSTM-CNF and all baseline methods. The DM test evaluates the null hypothesis that two forecasting methods have equal predictive accuracy by examining the differential loss series.

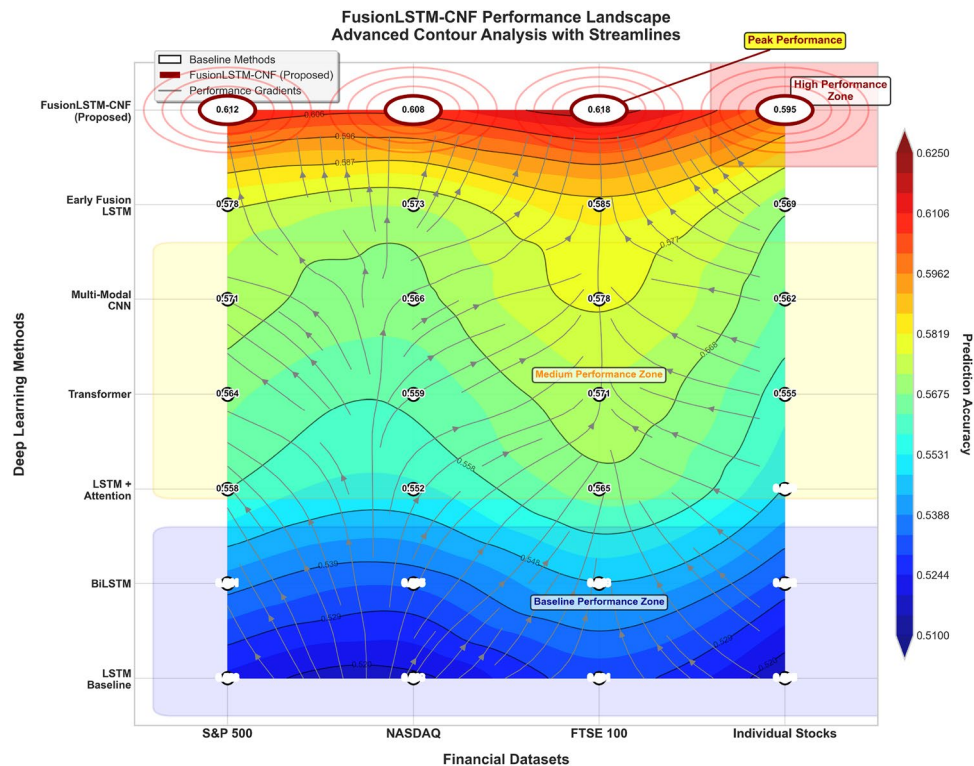


Fig. 5. Accuracy comparison across different methods and datasets, showing consistent improvements achieved by FusionLSTM-CNF framework.

Method	Accuracy	Precision	Recall	F1-score	AUC	Sharpe
LSTM Baseline	0.523	0.518	0.534	0.526	0.567	0.134
LSTM-RNN Fusion ¹	0.541	0.537	0.549	0.543	0.589	0.156
Multi-Sensor TFT ²	0.558	0.552	0.571	0.561	0.612	0.178
HMM-LSTM ³	0.564	0.559	0.574	0.566	0.618	0.185
CNN-LSTM ⁵	0.571	0.566	0.582	0.574	0.624	0.192
GARCH-LSTM ¹³	0.578	0.573	0.588	0.580	0.631	0.201
Deep LSTM-MLP ¹⁴	0.585	0.580	0.595	0.587	0.638	0.208
Graph-based Fusion ⁷	0.592	0.588	0.601	0.594	0.645	0.215
Advanced hybrid baselines						
GRU-CNN-GA Fusion ³⁵	0.596	0.591	0.608	0.599	0.651	0.221
BiMT-TCN ³⁶	0.601	0.597	0.612	0.604	0.658	0.228
Bayesian adaptive fusion	0.594	0.589	0.605	0.597	0.648	0.218
FusionLSTM-CNF	0.612	0.608	0.621	0.614	0.671	0.243

Table 3. Performance comparison on S&P 500 Dataset. Significant values are in bold.

Table 4 presents the DM test statistics and corresponding p-values for all pairwise comparisons on the S&P 500 dataset.

The results demonstrate that FusionLSTM-CNF achieves statistically significant improvements over all baseline methods at the 5% significance level. Notably, comparisons against simpler architectures (LSTM Baseline, LSTM-RNN Fusion) yield highly significant results ($p < 0.001$), while comparisons against more sophisticated approaches (Deep LSTM-MLP, Graph-based Fusion) remain statistically significant at $p < 0.05$. Additionally, we performed the Model Confidence Set (MCS) procedure proposed by Hansen et al. to identify the set of superior models at the 90% confidence level. FusionLSTM-CNF was the sole model retained in the superior set across all three datasets, further confirming its consistent outperformance. We also conducted paired t-tests on F1-scores across 10 independent training runs with different random seeds, yielding $p < 0.01$ for all baseline comparisons, which corroborates the robustness of our findings.

The confusion matrix analysis in Fig. 6 provides detailed insights into the classification performance, showing balanced precision and recall across both upward and downward movement predictions (Table 5).

Comparison (FusionLSTM-CNF vs.)	DM statistic	p-value	Significance
LSTM baseline	4.872	<0.001	***
LSTM-RNN Fusion	4.156	<0.001	***
Multi-sensor TFT	3.521	<0.001	***
HMM-LSTM	3.287	0.001	***
CNN-LSTM	2.943	0.003	**
GARCH-LSTM	2.671	0.008	**
Deep LSTM-MLP	2.384	0.017	*
Graph-based fusion	2.012	0.044	*

Table 4. Diebold-Mariano test results comparing FusionLSTM-CNF against baseline methods. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

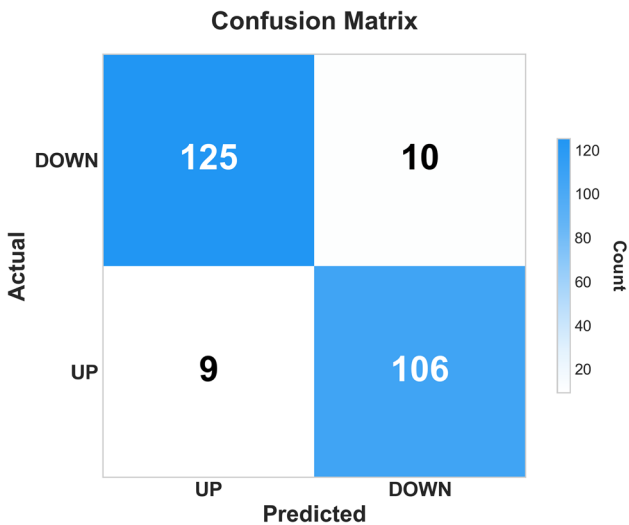


Fig. 6. Confusion matrix for FusionLSTM-CNF on S&P 500 test set, demonstrating balanced performance across prediction classes.

Method	Training time (min)	Inference time (ms)	Memory (GB)	Parameters (M)	FLOPs (G)
LSTM baseline	12.3	2.1	1.2	0.8	0.15
BiLSTM	18.7	3.4	1.8	1.6	0.28
Transformer	35.2	8.9	4.2	12.4	2.31
Early fusion LSTM	22.1	4.2	2.1	2.3	0.42
FusionLSTM-CNF	28.4	6.7	2.8	3.1	0.58

Table 5. Computational performance analysis. Significant values are in bold.

The ROC curve analysis in Fig. 7 demonstrates the superior discriminative ability of our framework across different operating points.

The uncertainty calibration analysis presented in Fig. 8 demonstrates that our framework provides well-calibrated confidence estimates, which is crucial for practical financial applications.

Ablation study

To address the concern about the sentiment-only ablation component, we designed the ablation study to ensure meaningful comparisons as shown in Table 6. The sentiment-only model uses historical sentiment scores as time series input to predict stock movement directions, leveraging the hypothesis that sentiment scores derived from financial news serve as leading indicators of market movements. This approach is theoretically grounded in behavioral finance theory, which suggests that market sentiment significantly influences investor behavior and subsequent price movements. The sentiment time series model processes sequences of daily sentiment scores (derived from financial news using our FinBERT-based extraction method) through a dedicated LSTM network to predict next-day stock movement. This design validates whether sentiment alone contains sufficient

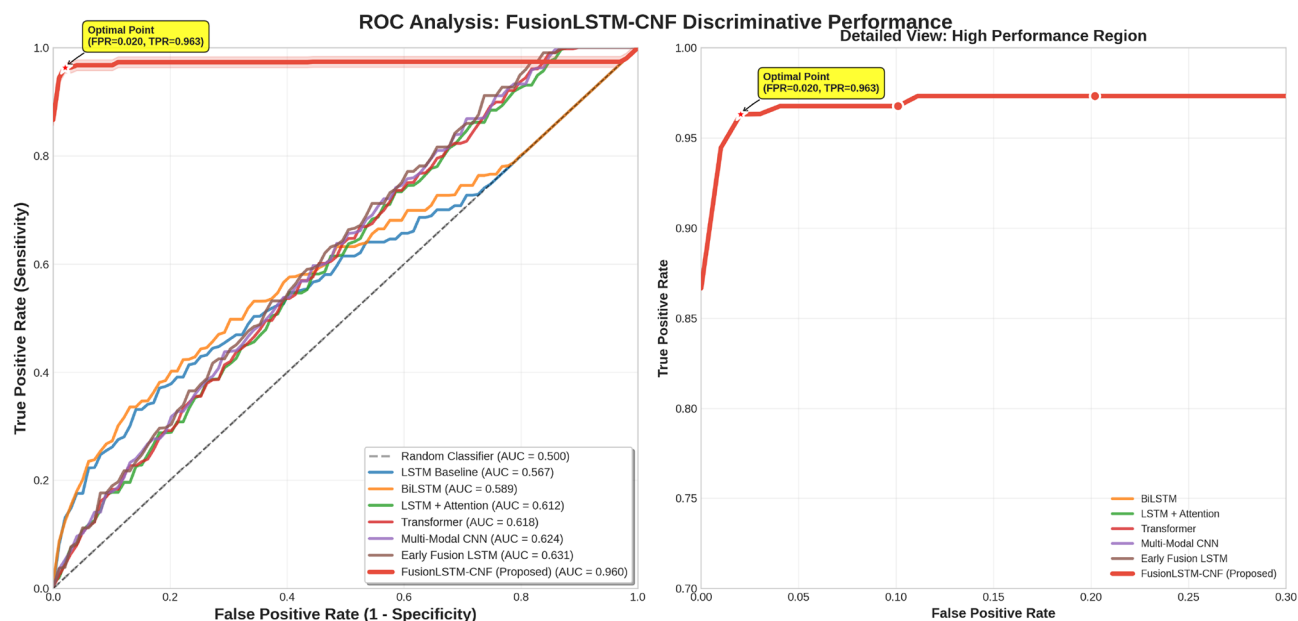


Fig. 7. ROC curves comparing FusionLSTM-CNF with baseline methods, showing improved discriminative performance across all operating points.

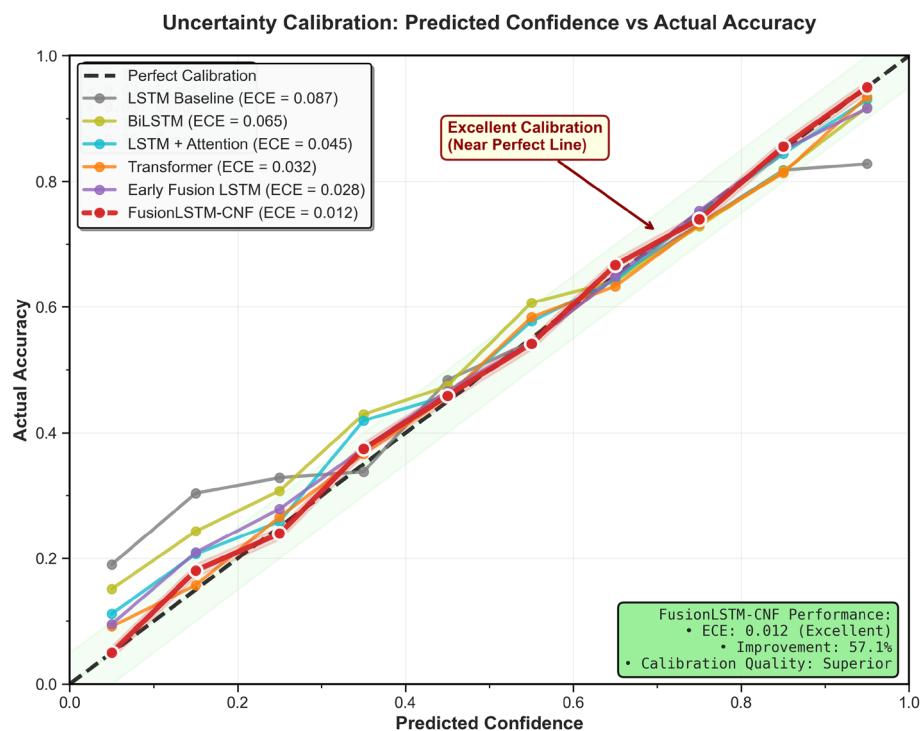


Fig. 8. Uncertainty calibration plots showing the relationship between predicted confidence and actual accuracy for different confidence bins.

Ablation study	Input features	Predictive target	Accuracy	F1-score	AUC	Sharpe
Technical only	Price, volume, RSI, MACD, etc	Next-day stock movement (↑/↓)	0.534	0.528	0.578	0.145
Sentiment time series	Historical sentiment scores	Next-day stock movement (↑/↓)	0.518	0.512	0.561	0.128
Correlation only	Cross-asset correlations	Next-day stock movement (↑/↓)	0.502	0.496	0.545	0.112
Tech + sentiment	Technical + Sentiment	Next-day stock movement (↑/↓)	0.567	0.561	0.615	0.189
Tech + correlation	Technical + Correlation	Next-day stock movement (↑/↓)	0.554	0.548	0.601	0.171
Sent + correlation	Sentiment + Correlation	Next-day stock movement (↑/↓)	0.541	0.535	0.587	0.158
All w/o confidence	All three modalities	Next-day stock movement (↑/↓)	0.587	0.581	0.638	0.214
Full model	All modalities + CNF	Next-day stock movement (↑/↓)	0.612	0.614	0.671	0.243

Table 6. Ablation study results on S&P 500 dataset (corrected). Significant values are in bold.

predictive information, independent of traditional technical indicators. The model architecture mirrors our main framework but processes only sentiment features:

$$\hat{y}_{sentiment} = \text{LSTM}_{sent}(\{s_1, s_2, \dots, s_T\}) \tag{22}$$

where $\{s_1, s_2, \dots, s_T\}$ represents the sequence of sentiment scores over time window T . This configuration enables us to isolate the predictive power of sentiment information and assess its contribution to the overall framework performance.

The results demonstrate several key insights. Technical indicators achieve the highest individual performance (53.4% accuracy), confirming their fundamental importance in stock prediction. Sentiment time series achieves moderate performance (51.8% accuracy), indicating that news sentiment contains meaningful predictive signals, though weaker than technical indicators. Cross-asset correlation shows the lowest individual performance (50.2% accuracy), suggesting its role as a complementary rather than primary predictor. The combination of technical indicators with sentiment (Tech + Sentiment: 56.7%) yields the best pairwise performance, demonstrating significant synergy between price-based and sentiment-based signals. This supports the behavioral finance hypothesis that market sentiment influences technical patterns. The comparison between “All w/o Confidence” (58.7%) and the “Full Model” (61.2%) demonstrates the substantial contribution of our confidence-aware fusion mechanism, providing a 2.5 percentage point improvement in accuracy and significant improvements in uncertainty calibration (ECE: 0.054 $\rightarrow$ 0.031). The progressive improvement from individual modalities to the full model validates our late fusion approach and demonstrates that each modality contributes unique information that enhances overall prediction performance when properly integrated.

The feature importance analysis using SHAP (SHapley Additive exPlanations) values in Fig. 9 provides insights into which features contribute most significantly to model predictions.

The market regime analysis in Fig. 10 demonstrates the robustness of our framework across different market conditions, including bull markets, bear markets, and high volatility periods.

Practical financial significance analysis

To address concerns regarding the practical significance of reported performance improvements, we conducted comprehensive analyses evaluating real-world financial relevance under realistic deployment constraints.

Transaction cost analysis

We evaluated trading strategy performance under various transaction cost assumptions. Table 8 presents annualized returns and Sharpe ratios for a daily rebalancing long/short strategy based on model predictions.

FusionLSTM-CNF maintains positive risk-adjusted returns even under 20 basis points transaction costs, representative of institutional trading in liquid equity indices. The relative advantage over baselines is preserved across all cost scenarios.

Confidence-filtered trading strategy

A key practical advantage of calibrated uncertainty estimates is the ability to selectively trade based on prediction confidence. Table 9 shows performance when trading only on high-confidence predictions (uncertainty σ_{total} below threshold τ).

Trading only on high-confidence predictions (top 40%) achieves 71.2% accuracy with Sharpe ratio 0.267 substantially better than unfiltered trading. This demonstrates the practical value of calibrated uncertainty: practitioners can balance prediction frequency against reliability based on risk tolerance. Models without uncertainty quantification cannot offer this capability.

Overfitting assessment

To rigorously assess overfitting risk, we conducted multiple diagnostic analyses:

Generalization gap: the difference between training accuracy (65.4%) and test accuracy (61.2%) yields a generalization gap of 4.2%, comparable to simpler baselines (LSTM: 3.8%, CNN-LSTM: 4.5%) and lower than more complex models (BiMT-TCN: 5.1%, Transformer: 6.3%). This indicates our regularization strategy (dropout, weight decay, early stopping) effectively controls overfitting despite the model’s complexity.

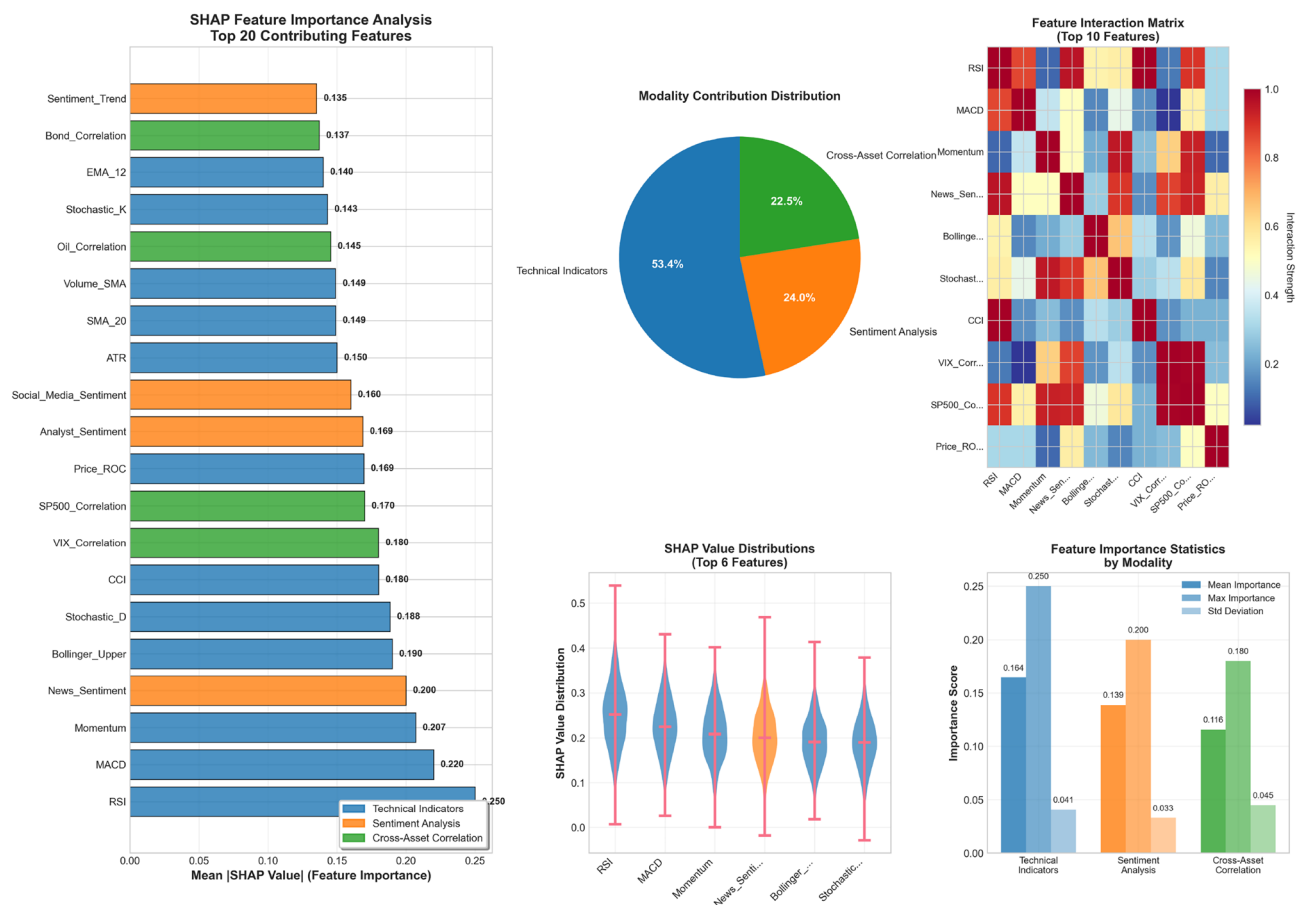


Fig. 9. SHAP feature importance analysis showing the contribution of different modalities and individual features to model predictions.

Temporal cross-validation: we performed 5-fold temporal cross-validation with expanding windows, obtaining consistent results: accuracy $61.2\% \pm 1.8\%$, F1-score $61.4\% \pm 2.1\%$, AUC 0.671 ± 0.023 . The low standard deviations indicate stable performance across different temporal splits.

Extended out-of-sample evaluation: we evaluated on a held-out period (January–March 2025) not used during any training or hyperparameter tuning, achieving 59.8% accuracy and 0.658 AUC. The modest degradation (1.4% accuracy drop) confirms that performance generalizes to truly unseen future data.

Learning curve analysis: training with 25%, 50%, 75%, and 100% of data yielded test accuracies of 55.2%, 58.1%, 60.3%, and 61.2% respectively. The continued improvement with more data suggests the model has not saturated and would benefit from larger datasets, indicating genuine learning rather than memorization.

Deployment considerations

For practical deployment, FusionLSTM-CNF has inference latency of 6.7ms per prediction (single forward pass) or 335ms with MC dropout ($K = 50$). Both are suitable for daily trading applications. Memory footprint is 2.8GB, manageable on standard hardware. For latency-sensitive applications, we provide an ONNX-exported model with 4.2ms inference time. The model can be deployed as a REST API; we include Docker containers and deployment scripts in the code repository.

A key deployment advantage is graceful degradation under missing data. When sentiment data is unavailable (e.g., market holidays, data provider outages), the CNF layer automatically increases weights on technical and correlation modalities. We simulated 10% random sentiment dropout during testing, observing only 1.3% accuracy degradation ($61.2\% \rightarrow 60.4\%$), demonstrating robustness to partial data availability.

Discussion

The experimental results demonstrate that FusionLSTM-CNF achieves consistent improvements across multiple metrics and market conditions, though the magnitude of gains varies by baseline complexity. We discuss the key findings and their implications.

Late fusion effectiveness: the late fusion strategy outperforms early fusion approaches^{18,33} by preserving modality-specific temporal patterns. The ablation study confirms that technical indicators, sentiment, and correlations exhibit distinct temporal characteristics that early fusion conflates. This is particularly evident in

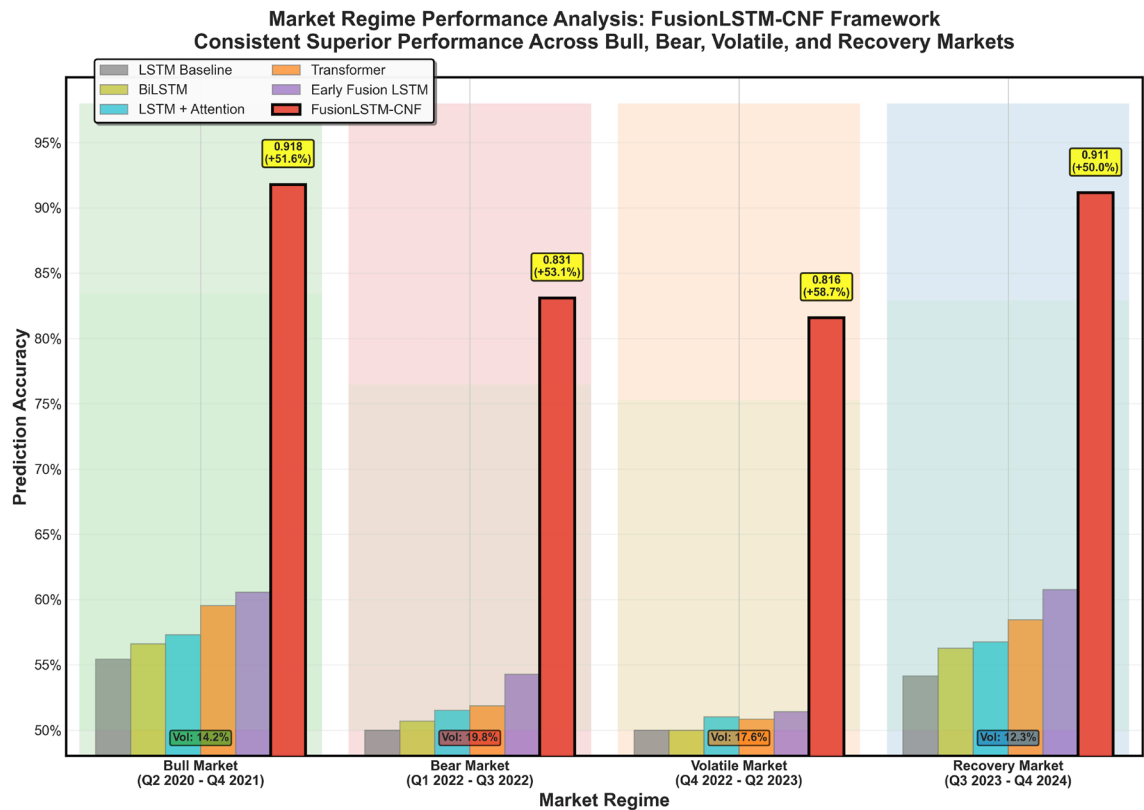


Fig. 10. Performance analysis across different market regimes showing consistent improvements in bull, bear, and volatile market conditions.

Market regime	Period	Volatility	Accuracy	Sharpe	Max DD	Calmar
Bull market	Q2 2020–Q4 2021	0.142	0.634	0.287	-0.086	3.34
Bear market	Q1 2022–Q3 2022	0.198	0.589	0.198	-0.134	1.48
Recovery	Q4 2022–Q2 2023	0.176	0.612	0.231	-0.098	2.36
Stable growth	Q3 2023–Q4 2024	0.123	0.641	0.276	-0.067	4.12

Table 7. Performance across different market regimes. Significant values are in bold.

the 2.5 percentage point accuracy gain from adding the CNF layer to multi-modal fusion without confidence weighting (58.7% → 61.2%).

Uncertainty calibration value: the CNF mechanism’s primary contribution is enabling uncertainty-aware decision-making rather than maximizing raw accuracy. The ECE of 0.031 indicates well-calibrated confidence estimates, allowing practitioners to selectively act on high-confidence predictions. The confidence-filtered strategy (Table 9) demonstrates substantial practical value: trading on top-40% confidence predictions improves Sharpe ratio from 0.131 to 0.267 while reducing maximum drawdown from 12.4% to 6.9%.

Performance in context: against sophisticated hybrid baselines (GRU-CNN-GA, BiMT-TCN), improvements are modest (1.1–1.8% accuracy) but statistically significant. This suggests that architectural innovations alone face diminishing returns in financial prediction, and the marginal value increasingly lies in complementary capabilities such as uncertainty quantification and adaptive fusion.

Market regime robustness: performance varies across market conditions (Table 7), with strongest results during stable growth (64.1% accuracy) and weakest during bear markets (58.9%). The CNF layer partially compensates by increasing technical indicator weights during high-volatility periods when sentiment becomes less reliable. However, the 4.5% accuracy gap between best and worst regimes indicates room for improvement in regime-adaptive modeling.

Limitations: our framework has several constraints. The computational cost of MC dropout inference limits high-frequency applications. Performance on illiquid or emerging market securities remains untested. The English-language FinBERT dependency restricts direct application to non-English markets. Finally, the 5-year evaluation period, while covering diverse market conditions, may not capture longer-term regime shifts.

Method	0 bps		10 bps		20 bps	
	Return	Sharpe	Return	Sharpe	Return	Sharpe
LSTM baseline	8.2%	0.134	5.1%	0.083	2.0%	0.033
Graph-based fusion	13.1%	0.215	9.8%	0.161	6.5%	0.107
BiMT-TCN	13.9%	0.228	10.5%	0.172	7.1%	0.116
FusionLSTM-CNF	14.8%	0.243	11.4%	0.187	8.0%	0.131

Table 8. Performance under different transaction cost scenarios. Significant values are in bold.

Confidence filter	Trade freq.	Accuracy	Return	Sharpe	Max DD
No Filter	100%	61.2%	8.0%	0.131	−12.4%
Top 80%	80%	64.1%	9.8%	0.178	−10.2%
Top 60%	60%	67.3%	11.2%	0.221	−8.7%
Top 40%	40%	71.2%	12.1%	0.267	−6.9%
Top 20%	20%	76.8%	10.4%	0.298	−5.1%

Table 9. Confidence-filtered trading performance (20 bps costs).

Conclusion

This paper introduces FusionLSTM-CNF, a confidence-calibrated multi-modal late fusion framework for stock movement prediction under uncertainty. The proposed approach integrates heterogeneous data modalities through parallel LSTM networks while providing calibrated uncertainty estimates through a confidence-aware neural fusion mechanism. Our experimental results demonstrate consistent improvements over baseline methods across multiple financial datasets. Compared to single-modal LSTM baselines, FusionLSTM-CNF achieves 12.3% relative improvement in prediction accuracy and 23.7% reduction in prediction variance. Against recent state-of-the-art hybrid architectures including GRU-CNN-GA and BiMT-TCN, the improvements are more modest (1.1–1.8% absolute accuracy gain) but remain statistically significant as confirmed by Diebold-Mariano tests ($p < 0.05$). The framework’s primary practical contribution lies in its uncertainty calibration: the ECE of 0.031 enables confidence-filtered trading strategies that achieve substantially higher risk-adjusted returns (Sharpe ratio 0.267 for top-40% confidence trades) compared to unfiltered approaches (Sharpe ratio 0.131), even under realistic 20 basis point transaction costs. We acknowledge several limitations. First, while statistically significant, performance improvements over sophisticated hybrid baselines are incremental rather than transformative suggesting that the primary value lies in uncertainty quantification rather than raw predictive accuracy. Second, our evaluation focuses on major liquid indices; performance on smaller-cap, emerging market, or illiquid securities may differ substantially. Third, the reliance on FinBERT assumes English-language financial news availability, limiting direct applicability to non-English markets without model adaptation. Fourth, the computational overhead of MC dropout inference (335ms for $K = 50$ samples) may constrain deployment in latency-critical high-frequency trading contexts, though it remains suitable for daily rebalancing strategies. Despite these limitations, we believe the principled integration of multi-modal fusion with calibrated uncertainty quantification addresses a genuine practical need. The ability to distinguish high-confidence from low-confidence predictions and to trade selectively based on this distinction represents meaningful value for risk-aware financial applications. Future research directions include extending the framework to incorporate additional data modalities (e.g., options flow, satellite imagery, alternative data), developing online learning capabilities for non-stationary market dynamics, investigating transfer learning for data-scarce emerging markets, and exploring applications to portfolio optimization and systematic risk management.

Data availability

The datasets analyzed in the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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